Rushabh Patel

Machine Learning and GenAI Leader

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SUMMARY

Nine-plus years building NLP, predictive analytics and LLM-powered systems in regulated, data-heavy industries. I turn business problems into production models, lead distributed teams, and care about governance, monitoring and responsible AI as much as accuracy.

SKILLS

GenAI & LLMs: RAG, Agentic workflows, Multi-agent systems, Prompt engineering, Few-shot learning, LLM evaluation, Fine-tuning (LoRA, QLoRA), LLM serving (vLLM, SGLang), Quantization (AWQ, GPTQ), LangChain, LangGraph, LlamaIndex, Generative AI, Vector databases, Responsible AI & guardrails

NLP: Named entity recognition, Text classification, Topic modeling, Clinical NLP, Speech to text, Parsing & POS tagging

Machine learning: Gradient boosting (XGBoost), Random forests, Deep learning, Computer vision & imaging, Time-series forecasting, Knowledge graphs & GNNs, Bayesian modeling & MCMC, Missing-data imputation, Clustering & segmentation, Clinical risk prediction, GANs, PyTorch, TensorFlow, AutoML (H2O)

MLOps & platforms: CI/CD for ML, AWS SageMaker, Apache Airflow, Feature stores (Feast), Docker & containers, Kubernetes, Argo Workflows, Dagster, Terraform, Model APIs, Model monitoring, Vertex AI

Cloud & data: AWS, Azure, Google Cloud, Snowflake, Apache Spark, Databricks, Apache Beam, Palantir, Tableau

Languages: Python, R, SQL, Julia, JavaScript & Node.js

EXPERIENCE

SVP, Gen AI Tech Lead

Citigroup | New York, NY | May 2025 – Present

- Leads a globally distributed team of GenAI developers, UI engineers and backend specialists in Citi's Controls Technology group, taking AI solutions from concept to production.
- Orchestrates cross-functional teams across geographies to deliver production-ready GenAI applications across the AI, UI and backend layers.
- Designs retrieval-augmented generation (RAG), agentic workflows and multi-agent systems for real-world risk and control use cases across the enterprise.
- Leads the Controls Tech GenAI roadmap: reviews new research, benchmarks emerging techniques and rapidly prototypes new architectures.
- Solves high-impact problems with NLP (NER, text classification, topic modeling) and LLMs through prompt engineering and RAG with deep retrieval strategies.
- Embeds responsible-AI principles and operational guardrails in every deployment, aligned with Citi's risk standards and enterprise controls.

Tools: Python, LangChain, LangGraph, AWS, Azure, Snowflake, Vector databases, Docker, Kubernetes

Lead Data Scientist (DS Manager II), GenAI

AIG | Atlanta, GA | Jan 2025 – May 2025

- Owned the full development and deployment lifecycle of data science and GenAI/ML models, from concept to operational rollout, within ethical and regulatory standards.
- Partnered with product managers, business leaders and engineers across the organization to build high-impact, data-driven solutions.
- Improved LLM performance on insurance-domain tasks with retrieval-augmented generation (RAG), prompt engineering and few-shot techniques.
- Built evaluation frameworks to measure LLM efficacy and ground-truth dataset quality, and used them to guide the product roadmap.
- Fostered a collaborative data science community built on knowledge sharing and creativity.

Tools: Python, LangChain, AWS SageMaker, Snowflake, SQL, Docker

Senior Machine Learning Engineer
VNS Health | New York, NY | Dec 2022 – Jan 2025

- Led the deployment of generative AI to transcribe, parse and structure call-center audio with open-source models, improving healthcare outcomes and customer service.
- Led the redesign of the ML architecture on AWS SageMaker, Apache Airflow and a Feast feature store for scalable, reliable infrastructure.
- Introduced a CI/CD-based architecture for ML that made model development faster and significantly cut the time to production.
- Coordinated cross-functional teams to establish ML best practices across the organization.
- Integrated state-of-the-art LLM techniques into AI-driven solutions.

Tools: Python, R, AWS SageMaker, Apache Airflow, Feast, Snowflake, LangChain, Node.js, JavaScript, SQL

Senior Machine Learning Engineer
EvolutionIQ | New York, NY | Oct 2022 – Dec 2022

- Built and deployed production ML models for claim time-series forecasting and NLP text analysis, improving predictive accuracy and processing efficiency.
- Ran exploratory analysis to find ML opportunities, diagnose data issues and set up model design, training and evaluation frameworks.
- Proposed and implemented new features that improved model performance and refined business logic.
- Wrote and peer-reviewed production code for robust, maintainable ML solutions.

Tools: Python, TensorFlow, PyTorch, Dagster, BigQuery, Vertex AI, GCP, Terraform, Spark, Kubernetes

Senior Data Scientist
AT&T | New York, NY | Jun 2022 – Oct 2022

- Worked with industry partners at the intersection of databases, knowledge graphs and artificial intelligence.
- Designed and built AI and ML applications on knowledge-graph technology to solve large-scale, real-world problems.

Tools: Python, Azure Databricks, PySpark, H2O, Palantir, REL, Julia, SQL

Applied Data Scientist II
Children's Hospital of Philadelphia | Philadelphia, PA | Jun 2020 – Jun 2022

- Used natural language processing to turn clinical text into structured information.
- Classified imaging studies with deep learning and transfer learning.
- Applied Bayesian statistical models to incomplete or missing patient data.
- Built ML models and cloud data pipelines, and adapted published models to imperfect clinical data, including building training datasets.
- Wrote secure code that serves models as APIs and services, scaled with containers and cloud-managed services.

Tools: Python, R, Apache Spark, BigQuery, Dataflow, Apache Beam, Argo Workflows, Kubernetes, Terraform, Docker

Data Scientist
VNS Health | New York, NY | Feb 2019 – Jun 2020

- Built, tested and deployed ML models for a healthcare organization and its health plan, improving business outcomes and quality of care.
- Created and maintained a framework for deploying models through APIs.
- Monitored deployed models, alerted management when performance declined and found the causes.
- Found patient sub-populations that benefit from targeted care management, improving positive predictive value with a sub-model per cluster.
- Kept data quality high from sourcing and ground-truth generation through normalization and transformation.

Tools: Python, R, SQL, Apache Spark, Tableau, H2O

Data Scientist
Jvion | Atlanta, GA | Jun 2017 – Jan 2019

- Improved patient outcomes with statistical models in R and Python for healthcare research.
- Designed algorithms that recommend actions and interventions to prevent adverse events and deterioration.
- Modeled hospital readmission among AMI patients, inpatient and ER visits, MRSA among diabetes patients, congestive heart failure, C. diff, pressure injury, sepsis and fall injury.

- Clustered complex patient populations for targeted care management, with a sub-model per cluster.
- Wrote complex SQL for data investigation, and built algorithms for population-level insights and the expected ROI of predictive analytics.

Tools: Python, R, SQL, Apache Spark, Tableau, H2O

Co-founder
PayTooth Software Solutions | Pune, India | Jul 2015 – Aug 2016

- Proposed and built a way to transfer money when the cellular network is weak or missing.
- Designed and developed the PayTooth application, and published the system in the International Journal of Computer Applications.

Tools: Python, R

Undergraduate Intern
Allstate Solutions | Bangalore, India | Jan 2014 – May 2014

- Worked with the team behind the Document Processing System, which generates correspondence for everyone involved in a claim.
- Designed correspondence templates from business requirements.

EDUCATION

Ph.D., Computer Science
Auburn University | Auburn, AL | 2020 – 2026

- Pursued while working full time in industry.
- Began the program at Temple University (2020 – 2022) before continuing at Auburn.
- Research in graph machine learning and protein language models (see publications).

M.S., Computer Science
University of Illinois Springfield | Springfield, IL | 2016 – 2017

- GPA 4.0, with a concentration in data science.
- Honors award from the College of Liberal Arts & Sciences for academic performance.
- Excellence award for healthcare analytics research with SIU School of Medicine.
- Trained convolutional networks on breast-cancer imaging, presented at SIIM 2017.

B.Tech, Computer Science
Symbiosis Institute of Technology, Symbiosis International University | Pune, India | 2011 – 2015

- Published five papers on mobile payments, mobile networks and cloud systems before graduating.

PUBLICATIONS

Patel, R., Guo, Y., Alhudhaif, A., Alenezi, F., Althubiti, S. A., & Polat, K. (2022). Graph-based link prediction between human phenotypes and genes. Mathematical Problems in Engineering.

Buxton, E. K., Patel, R., Toumaian, M. R., & Raeeszadeh-Sarmazdeh, M. (2021). Application of protein language models to low-N engineering of metalloproteinase inhibitors. Intl. Conference on Computational Science and Computational Intelligence (CSCI).

Buxton, E. K., Vohra, S., Guo, Y., Fogleman, A., & Patel, R. (2019). Pediatric population health analysis of southern and central Illinois region: a cross-sectional retrospective study using association rule mining and multiple logistic regression. Computer Methods and Programs in Biomedicine.

Vespa, L. J., Guo, Y., Der, M., Patel, R., & Pan, X. (2017). A novel reinforcement sample learning strategy for convolution neural network in computer-aided diagnosis system for breast cancer. Society for Imaging Informatics in Medicine (SIIM).

PayTooth: a cashless mobile payment system based on Bluetooth. Intl. Journal of Computer Applications (2015).

A cloud-based public transport system. Intl. Journal of Computational Intelligence Research (2015).

A comparative review of existing mobile payment systems. Intl. Journal of Applied Engineering Research (2015).

I-RS: a railway system based on cloud computing. IJSETR (2015).

Evolution and envisage of mobile networks. IJSETR (2014).